BIKE SHARING DATASET

RAW FILES :

DAYWISE DATA :

HOURWISE DATA :

DAY WISE DATA :

day.csv

HOUR WISE DATA :

hour.csv

🚲 Bike Sharing Demand Analysis & Prediction

📌 Project Overview

This project analyzes bike-sharing demand using historical data and builds machine learning models to:

- Predict **hourly bike demand**
- Identify **peak rental hours**
- Understand **weather and temporal impacts**
- Provide **business-driven insights** for revenue optimization

The project follows a **complete data science lifecycle**, from EDA to model deployment thinking.

🎯 Business Problem

Bike-sharing companies need to:

- Allocate bikes efficiently
- Identify peak demand hours
- Reduce losses during low-demand periods
- Adjust operations based on weather & seasonality

This project helps answer:

> When should more bikes be deployed, and under what conditions does demand peak or drop?

📂 Dataset Description

The dataset contains **two CSV files**:

1️⃣ Day-wise Data (`day.csv`)

- **731 rows × 16 columns**
- Aggregated daily bike rentals

2️⃣ Hour-wise Data (`hour.csv`)

- **17,379 rows × 17 columns**
- Detailed hourly rental information

Target Variable

- `cnt` → Total number of bike rentals

🧹 Data Preprocessing

Key preprocessing steps:

- Dropped non-informative columns (`dteday`, `instant`)

- Removed **data leakage features** (`casual`, `registered`)
- Selected `atemp` over `temp` (feels-like temperature)
- Scaled numerical features using `StandardScaler`
- One-hot encoded categorical features where required

📊 Exploratory Data Analysis (EDA)

Key Observations

- Strong correlation with demand:
  - `hr`, `workingday`, `season`, `temp/atemp`
- Weak or negative correlation:
  - `holiday`, `humidity`
- Demand peaks during:
  - Morning commute (7–9 AM)
  - Evening commute (5–7 PM)
- Weather impacts demand but **does not dominate it**

🛠️ Feature Engineering

- Created **peak hour labels** (without data leakage)
- Encoded categorical variables
- Removed features unavailable at prediction time
- Ensured realistic real-world prediction setup

🤖 Models Trained

🔹 Hourly Demand Prediction (Regression)

Model	R^2	MAE	RMSE
Linear Regression	0.65	81.36	105.75
Ridge Regression	0.65	81.36	105.75

Model	R^2	MAE	RMSE
Random Forest	**0.95**	**24.15**	**40.06**

✅ **Random Forest outperformed due to non-linear demand patterns**

🔹 Peak Hour Prediction (Classification – Leakage Free)

Metric	Value
Accuracy	**94%**
Precision (Peak)	0.89
Recall (Peak)	0.86
F1-score (Peak)	0.88

Confusion Matrix:

```
[[2550   86]
 [ 119  721]]
```

📌 This model is **realistic and deployable.**

🌦️ Weather Impact Analysis

With Weather Features

- Weather alone has **limited predictive power**
- Linear models performed better than tree models

Without Weather Features

- Significant performance improvement
- Indicates **time & human behavior dominate demand**

📌 **Insight**: Weather modifies demand but does not drive it.

🔍 Model Explainability

Feature Importance (Random Forest)

Top drivers:

- Hour of day (`hr`)
- Working day
- Season
- Temperature (secondary)
- Weather conditions

SHAP Analysis

- Confirms non-linear influence of time-based features
- Validates business intuition with model reasoning

🚀 Deployment Thinking

Prediction Pipeline

1. Input features (hour, season, weather forecast)
2. Apply same preprocessing
3. Predict demand
4. Trigger business actions

Example Business Actions

Prediction	Action
High demand	Increase bikes & pricing
Low demand	Reduce fleet
Bad weather	Shift bikes to shelters
Availability	Make bikes available during peak hours

📈 Model Monitoring (Future Scope)

- Track RMSE weekly
- Detect data drift

- Retrain models periodically
- Integrate real-time weather APIs

🧠 Key Learnings

- Data leakage can lead to misleadingly perfect models
- Strong metrics ≠ good model unless validated properly
- Data science is **analysis + reasoning + decisions**, not just models

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🔮 Future Work

- Time-series forecasting (LSTM / ARIMA)
- Real-time demand prediction
- Pricing optimization models

🏁 Conclusion

This project demonstrates a full data science workflow with:

- Strong EDA
- Correct modeling choices
- Explainable insights
- Business-focused recommendations

👨‍💻 Author

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BASIC INFORMTION ABOUT DATASET :

- `instant`: record index
- `dteday` : date
- `season` : season (1:springer, 2:summer, 3:fall, 4:winter)
- `yr` : year (0: 2011, 1:2012)
- `mnth` : month (1 to 12)
- `hr` : hour (0 to 23)
- `holiday` : weather day is holiday or not
- `weekday` : day of the week
- `workingday` : if day is neither weekend nor holiday is 1, otherwise is 0.

+ `weathersit` :
  - 1: Clear, Few clouds, Partly cloudy, Partly cloudy
  - 2: Mist + Cloudy, Mist + Broken clouds, Mist + Few clouds, Mist
  - 3: Light Snow, Light Rain + Thunderstorm + Scattered clouds, Light Rain + Scattered clouds
  - 4: Heavy Rain + Ice Pallets + Thunderstorm + Mist, Snow + Fog

- `temp` : Normalized temperature in Celsius. The values are divided to 41 (max)
- `atemp` : Normalized feeling temperature in Celsius. The values are divided to 50 (max)
- `hum` : Normalized humidity. The values are divided to 100 (max)
- `windspeed` : Normalized wind speed. The values are divided to 67 (max)
- `casual` : count of casual users
- `registered` : count of registered users
- `cnt` : count of total rental bikes including both casual and registered

SOME PROFFESIONAL LEVEL INSIGHTS :

- Hourly data enables operational decision-making by capturing intra-day demand patterns through the `hr` feature. It allows us to identify peak hours, demand drops, and behavior differences between working days and weekends, which daily data cannot reveal.
- Bike rentals show a strong bimodal daily pattern, with a morning peak around 8 AM and an evening peak between 5–6 PM, indicating usage driven primarily by office commuting.
- Working days exhibit sharp morning and evening demand peaks, while non-working days show a flatter demand distribution, indicating leisure-oriented usage rather than commuting.
- While seasonal demand levels vary significantly, the intra-day rental pattern remains consistent, suggesting that hour-based operational planning is stable year-round, with seasonal scaling required.

MODEL BUILDING STEPS :

Peak Hour Model

The Random Forest achieved an accuracy of 94% and an F1-score of 0.88 for peak hours. The model demonstrates strong recall (86%) for identifying high-demand periods, making it suitable for operational decision-making such as dynamic pricing and resource allocation.

Weather Impact

Weather variables alone were insufficient to predict demand. Removing weather features improved performance, indicating that temporal and behavioral factors dominate rental patterns.

Hour of day has the strongest positive impact on bike demand, especially during commute hours. Weather conditions moderately affect demand, but do not dominate predictions, confirming that behavioral patterns are the primary driver

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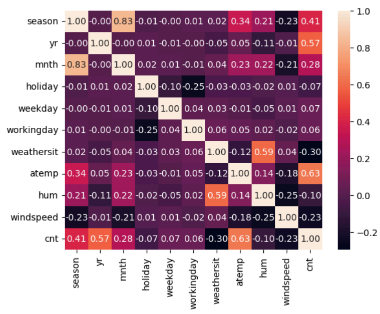
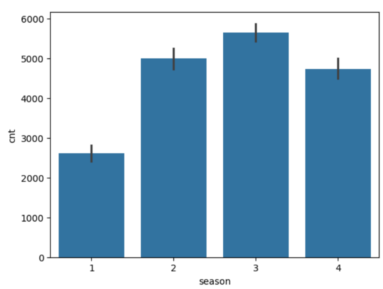
VISUALIZATIONS :

Day wise dataset

```
corr = data.corr()
sb.heatmap(corr, annot=True, fmt='.2f')
```

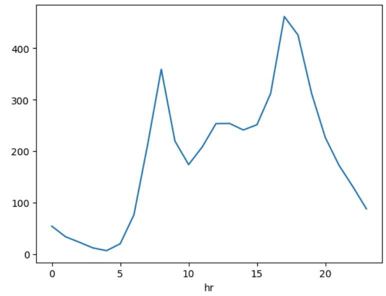



```
sb.barplot(x='season', y='cnt', data=data)
```

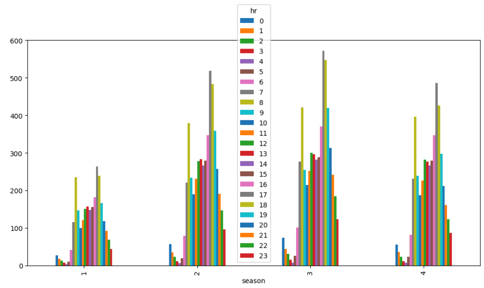


Hour wise dataset

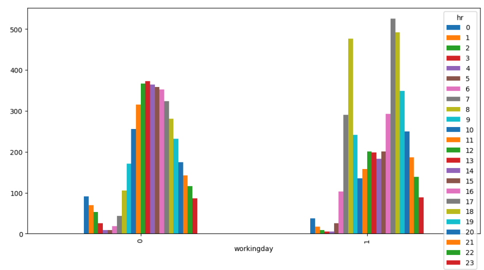
```
data.groupby('hr').mean()['cnt'].plot()
```



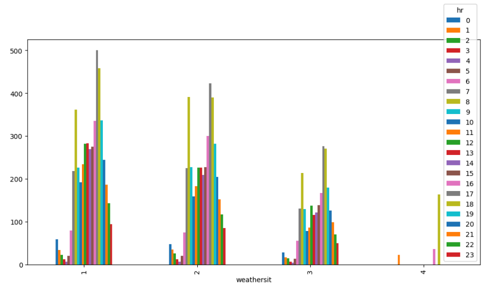

```
hourly_demand = data.groupby(['season', 'hr'])['cnt'].mean().unstack()
hourly_demand.plot(kind='bar', figsize=(12,6))
```




```
hourly_demand_workingdays = data.groupby(['workingday', 'hr'])['cnt'].me
an().unstack()
hourly_demand_workingdays.plot(kind='bar', figsize=(12,6))
```



```
hourly_demand_weather = data.groupby(['weathersit', 'hr'])['cnt'].mean().unstack()
hourly_demand_weather.plot(kind='bar', figsize=(12,6))
```



FIRST SIGHT INSIGHTS :

DATA HAS SOME FOLLOWING FEATURES :

- day wise data has 731 rows and 16 columns.
- I dropped `dteday` column because it has no role in our data.
- target variable is `cnt` (count of rental bikes per day
- relation of `cnt` with other features :

  `registered` 0.95
  `casual` 0.67
  `temp` 0.63
  `attemp` 0.63
  `yr` 0.57
  `season` 0.41
  `mnth` 0.28
  `weekday` 0.07
  `workingday` 0.06
  `holiday` -0.07
  `hum` -0.10
  `windspeed` -0.23
  `weathersit` -0.30

- NEGATIVE CORRELATION :: `hum` , `holiday` (very low)
- WEAK RELATION WITH TARGET VARIABLE :: `weekday` , `workingday`
- MEDIUM CORELATION :: `mnth` , `weathersit` , `windspeed`
- STRONG RELATION :: `yr` , `season` , `temp` , `attemp` , `casual` , `registered`

when `weathersit` is 1 on working day , `cnt` is 156
when `weathersit` is 2 on working day , `cnt` is 70
when `weathersit` is 3 on working day , `cnt` is 5
when `weathersit` is 1 on non-working day , `cnt` is 307
when `weathersit` is 2 on working day , `cnt` is 177
when `weathersit` is 3 on working day , `cnt` is 16

when `weathersit` is 1 on holiday , `cnt` is 448
when `weathersit` is 2 on holiday , `cnt` is 241
when `weathersit` is 3 on holiday , `cnt` is 21
when `weathersit` is 1 on non-holiday , `cnt` is 15
when `weathersit` is 2 on non-holiday , `cnt` is 6
when `weathersit` is 3 on non-holiday , `cnt` is 0

- dropping rows `registered` and `casual` because they are directly related to `cnt` and can cause data leakage.
- dropping `temp` and keeping `atemp` because For demand prediction, how temperature feels to humans matters more than actual temperature.

> 👉 - `temp` : Normalized temperature in Celsius. The values are divided to 41 (max)
> - `atemp` : Normalized feeling temperature in Celsius. The values are divided to 50 (max)

- Hour wise data has 17,379 rows and 17 columns

- average rentals by hour:

HOUR	Average rentals
0	53.898072
1	33.375691
2	22.869930
3	11.727403
4	6.352941
5	19.889819
6	76.044138
7	212.064649
8	359.011004
9	219.309491
10	173.668501
11	208.143054
12	253.315934
13	253.661180
14	240.949246
15	251.233196
16	311.983562
17	461.452055
18	425.510989
19	311.523352
20	226.030220
21	172.314560
22	131.335165
23	87.831044